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Semi-Autonomous Control System Helps Improve Coiled Tubing String Life. A Successful Case Study from Colombia.

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Abstract

This paper presents the implementation of a semi-autonomous control system designed to optimize Coiled Tubing (CT) operations by extending CT string service life, a critical objective in reducing operational costs and improving safety. The system determines appropriate injector traction and reel tension forces based on real-time physics-based models, which are continuously updated using surface sensor inputs, calculated parameters and tubing mechanical properties. This application was field tested in the Llanos Orientales area of Meta, Colombia.

The control system integrates electro-hydraulic valves with a Data Acquisition System (DAS) and programmable controller, enabling dynamic computation and application of control setpoints for the injector traction and reel hydraulic circuits. These setpoints are derived from continuously solved physics-based models that account for tubing wall thickness, axial loading, internal and external CT pressure, depth and other sensed/mechanical factors. The field trial involved mechanical cleaning of the well using a rotary tool in the production tubing with an electric submersible pump. The control system was activated while executed CT Running in Hole (RIH) and Pulling Out of Hole (POOH) during access of the 3 1/2 in tubing with 1.75 in CT from 150 ft to a target depth of 7700 ft.

During RIH, the system automatically applied 800 psi to the traction mechanism and continuously adjusted the injector traction force based on real-time model outputs, including during a pull test. The system also dynamically modulated reel pressures by adjusting from 700 psi to 1200 psi in response to variations in tubing wall thickness, weight, and changes in direction from RIH to POOH. The system satisfactorily assisted the CT operator by notifying adjustments in real-time, according to CT parameters and sensor readings, without affecting the operator's autonomy to decide whether to operate in automatic or manual mode.

System automation allows the application of correct traction and tension forces during CT operations by monitoring CT parameters, surface sensors, and controlling hydraulic pressures exerted by the injector and reel, taking into account the correlation of the CT region in which the axial, radial, and tangential stresses represented in the equivalent Von Mises stress equal material uniaxial resistance. This approach represents an advancement in CT automation, delivering practical value through model-driven force application, thereby extending CT life.

Introduction

CT has become an essential solution for well intervention. Its ability to perform continuous operations in live wells without the need for rig-based interventions makes it a preferred method for tasks such as cleanouts, logging, perforating, and deploying tools in complex well geometries among others. The inherent flexibility and cost-effectiveness of CT operations have led to widespread adoption, improving efficiency and safety operations.

Despite its advantages, CT operations are mechanically demanding. The string is subjected to repeated bending, axial tension and compression, differential pressures, and frictional contact with wellbore surfaces. These stress factors contribute to cumulative fatigue, wall thinning, and eventual structural failure. The challenge is compounded by the dynamic nature of well conditions during job execution and the variability in tubing properties along its length. As a result, maintaining the mechanical integrity of the CT string throughout its service life is a critical concern for operators and service providers.

Traditionally, CT units rely on manual control of hydraulic systems to regulate injector traction and reel tension forces. Operators use visual indicators, experience, tables, and surface feedback to make calculations and take real-time decisions while executing the planned operation which itself can vary on complexity depending on its nature therefore involving more attention from the operator. While this approach has proven effective in many scenarios, it is inherently reactive and limited in its ability to continuously adapt to changing mechanical and operational conditions. Human judgment, while valuable, is affected by subjectivity, experience and may not consistently account for complex stress interactions or subtle changes in tubing behavior.

Recent advances in automation, control application, and real-time modeling have opened new possibilities for improving CT reliability and performance. By integrating surface sensor data with physics-based models and programmable control systems, it is now feasible to develop intelligent systems capable of dynamically optimizing force application and CT hydraulic pressures control. These systems can monitor tubing stress states, predict fatigue accumulation, and adjust hydraulic pressure force parameters to maintain safe operating conditions—all in real time.

This paper introduces a semi-autonomous control system designed to address these challenges by extending CT string life and improving operational safety. The system combines real-time physics-based modeling with electro-hydraulic control to automatically regulate injector traction and reel tension forces. It continuously evaluates tubing stress conditions—including axial, radial, and tangential components—and correlates them with the Von Mises equivalent stress to ensure that applied forces remain within the material's uniaxial resistance limits across the entire CT depth profile. This approach not only improves operator decision-making but also ensures that mechanical forces remain within safe limits, thereby preserving equipment integrity and reducing downtime due to CT unexpected mechanical failures.

This paper builds on these motivations by presenting a field-tested solution that leverages real-time modeling and electro-hydraulic control to dynamically manage CT injector gripper and reel tension forces. The system's ability to adapt to changing conditions, provide actionable feedback, and automate control while preserving manual override capabilities represents a significant step forward in the evolution of CT automation.

System Architecture

The semi-autonomous control system developed for CT operations is designed to integrate real-time data acquisition, physics-based modeling, electro-hydraulic actuation and dynamic variables control into a cohesive control framework. The architecture allows adaptability to different CT strings and operational scenarios, while maintaining a focus on safety, reliability, and operator support.

System Overview

The control system consists of three primary subsystems:

- **Data Acquisition System (DAS):** Collects real-time data from surface sensors, including CT weight, internal pressure, external pressure, CT depth, injector speed and pressures from injector and reel hydraulic circuits that will work as control loop feedback.
- **Control and Modeling Unit:** A programmable logic controller continuously solves physics-based models to estimate stress conditions across the CT depth profile and determine optimal traction (gripper) and reel tension setpoints from all sensor data inputs and calculated data.
- **Electro-Hydraulic Actuation System:** Applies and controls the computed setpoints to the injector and reel hydraulic pressure circuits using proportional control valves.

In Fig. 1 and Fig. 2, we can see a high-level overview of the system architecture for both Gripper and Reel control systems.

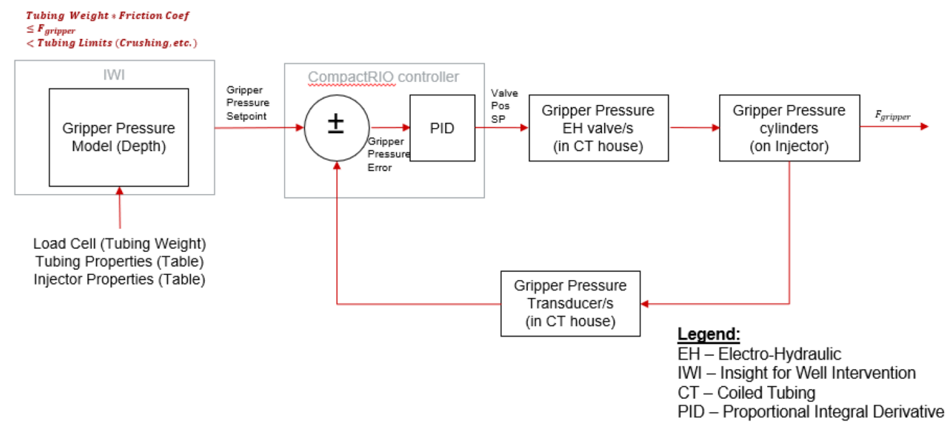


Figure 1—Gripper Control Loop

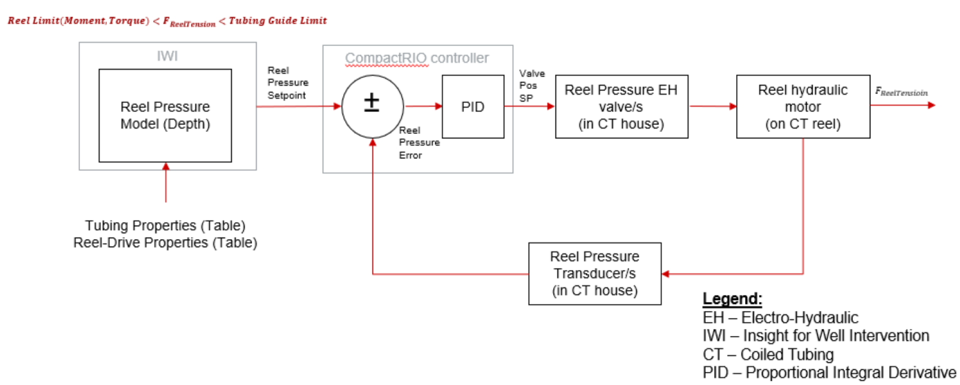


Figure 2—Reel Control Loop

Real-Time Modeling Engine

At the core of the system is a real-time modeling engine that continuously evaluates the mechanical state of the CT string. The model incorporates:

- Axial, radial, and tangential stress calculations.
- Internal and external pressures, as well as bending radius effects.
- CT depth profile material properties and corresponding fatigue thresholds.
- Von Mises equivalent stress to assess proximity to yield limits.

This model enables the system to predict the optimal force application required according to dynamic conditions of the operation and well conditions to minimize stress accumulation at any given CT depth and extend CT life.

Control Logic and Setpoint Generation

Based on model outputs, sensors data and corresponding calculate inputs, the programmable controller dynamically adjusts hydraulic pressures setpoints for:

- **Injector traction force**, ensuring necessary gripping force to safely hold the tubing, according to actual conditions. In this manner for example, when CT high weights are involved, the gripping force is adequate to prevent CT string slipping or run away. Similarly, when the CT weight decreases, the gripping force decreases accordingly, avoiding the CT string from being exposed to unnecessary gripping force.
- **Reel tension pressure**, maintaining controlled spooling and unspooling of the CT string, according to parameters like, injector direction, CT string wall thickness and calculated parameters.

The control logic includes safety thresholds, hysteresis bands, and operator override capabilities to ensure robust and fail-safe operation, without preventing the operator from taking control at any time during operation. Hydraulic pressures setpoints are always generated and displayed to the operator no matter if the control system is in automatic or manual mode.

Operator Interface

A Human-Machine Interface (HMI) displays real-time system status, sensor readings, calculate parameters and model outputs. Operators can switch between manual and automatic modes at any time during a giving operation execution point, receive alerts for abnormal conditions, and adjust control parameters within predefined safety limits.

Operator Interaction and Override

While the system operates autonomously, it is designed to support—not replace—the CT operator. The Human-Machine Interface (HMI) provides:

- Real-time visualization of stress levels, controlled hydraulic pressures, CT operational variables.
- Notifications when model outputs suggest adjustments in automatic or manual operation modes.
- Manual override capability, allowing the operator to switch between automatic and manual modes at any time without compromising the safe course of the operation.
- Independent injector gripper and reel tension hydraulic control circuits allowing the operator to decide whether to activate both systems or either one of them, for the duration of the operation established period.

This hybrid approach ensures that automation enhances decision-making without compromising operator control or operation safety.

Case History

While the control system has not been activated, the system can be used as an advisory. The operator can view suggested setpoints for gripper and reel hydraulic circuits on a numeric display.

Calc Gripper Op Pr Setpnt	810	psi
Calc Gripper Max Pr Stpnt	1200	psi
Calc Gripper Press Max	2370	psi
Calc Gripper Press Min	510	psi
Calc Reel Pressure Setpnt	650	psi
Calc Reel Pressure Max	2750	psi
Calc Reel Pressure Min	500	psi

Figure 2a—Control System Advisory Display

Before first activation in field, control system calculations were tracked and followed in previous jobs to validate that the suggested setpoints were feasible, they were compared with the values used during operation. As we can see in Fig. 3, Calc Gripper Op Pr Setpnt (800 psi), was similar to manual applied gripper pressure observed in Lower and Upper Gripper Pressure (900 psi). It was also noticed that at target Depth (8000 ft), when CT weight increased to around 20000 lbf, the Cal Gripper Op Pr Setpnt increased for those moments.

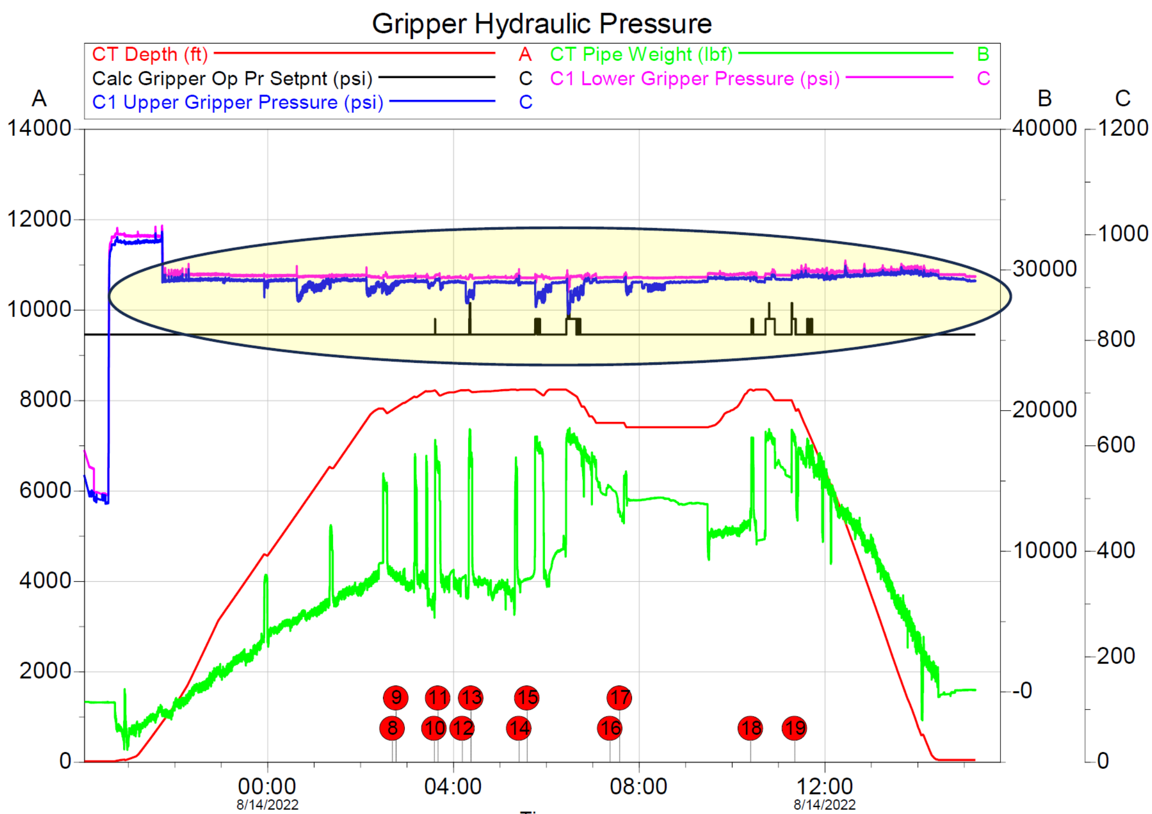


Figure 3—Control System Gripper Track

The same monitoring was done on reel pressures. In Fig 4. We could observe that it was necessary to apply more Reel Back Tension pressure (600 psi) than the suggested by Calc Reel Pressure Setpoint (400 psi), because operator observed that the tubing began to slack in the reel. This was corrected in the control system for field execution.

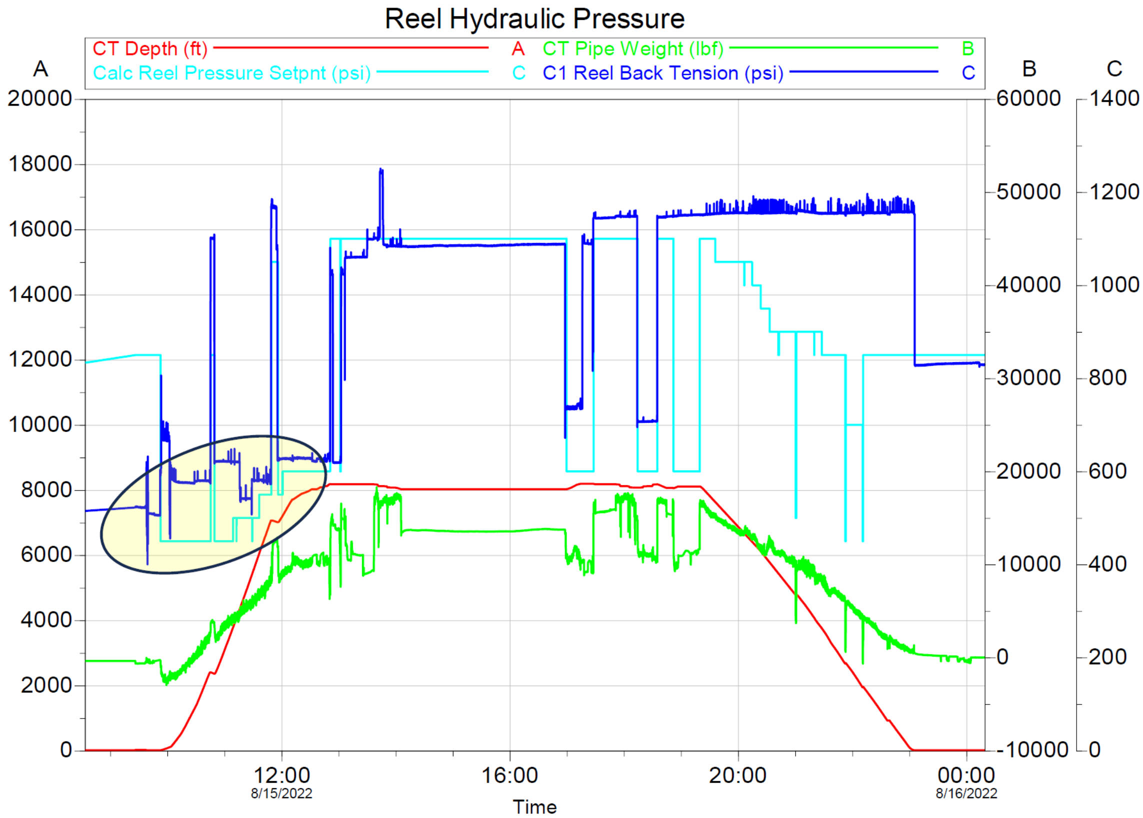


Figure 4—Control System Reel Track

Field Implementation and Results

To validate the performance and reliability of the semi-autonomous control system, a field trial was conducted in the Llanos Orientales region of Meta, Colombia. The operation involved mechanical cleaning of a customer production well using a rotary tool deployed via a 1.75-inch CT string inside 3½-inch production tubing. The intervention reached a total measured depth of 7,700 feet and was performed with an electric submersible pump (ESP) in place, inside completion well.

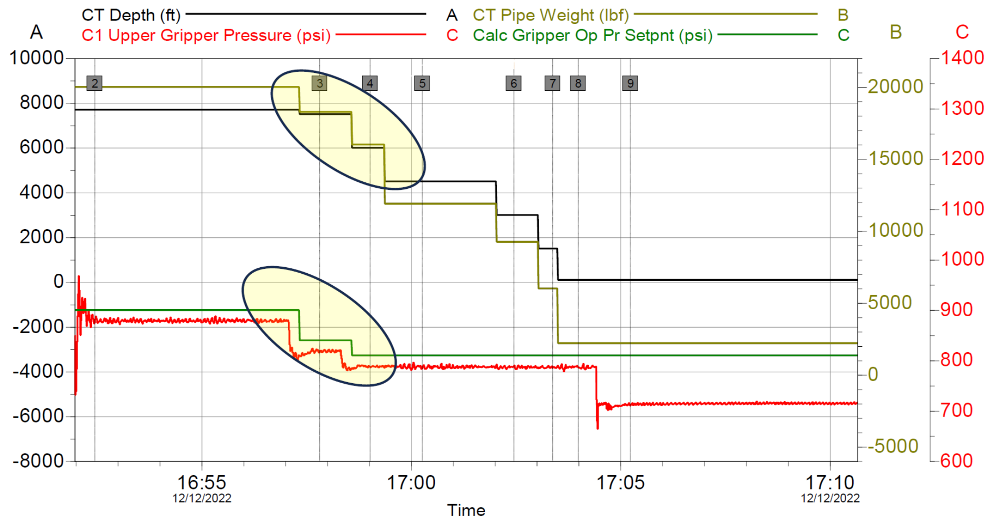
Operational Setup

The control system was installed on a CT unit equipped with:

- A programmable controller for dynamic computation.
- Electro-hydraulic valves for injector and reel pressure control.
- Data Acquisition System for surface sensors reading, internal pressure, external pressure, string depth, string weight, injector speed, pressures from injector and reel hydraulic circuits.
- A Human-Machine Interface (HMI) for operator interaction and system monitoring.

The system was tested before job execution, simulating well conditions and evaluating the system's good performance. CT weight during RIH and POOH simulation and corresponding depths were used as the control system input so programmable logic controller could generate corresponding control output variables to validate the correct hydraulic system control response and programmable controller correct calculated variables.

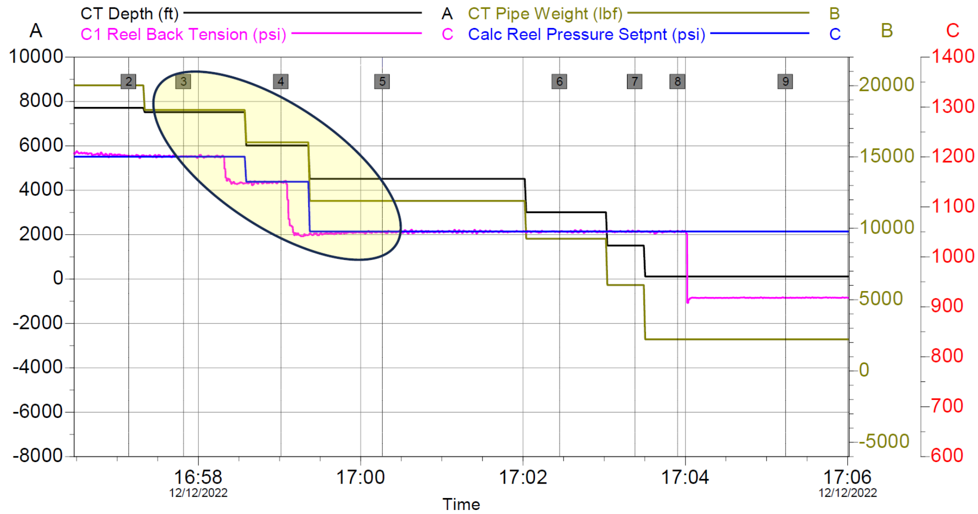
With the control system activated for both injector gripper and reel tension hydraulic control circuits, the expected values of weight and depth from bottom hole to surface were entered to validate the dynamic setting and hydraulic gripper control pressure system performance of the injector traction force identified as Gripper Pressure. We observed that as the weight decreases, in this case, the Gripper Pressure set point decreases, and the corresponding adjustment of the pressure applied in the hydraulic system was achieved and controlled, evidenced in events 3 and 4, as illustrated in Fig. 5.



	Intersection		CD	CPW	CUGP	CGOPS		Intersection		CD	CPW	CUGP	CGOPS
2	Copilot Enabled	16:52:27	7700	20000	875.2	900.0	3	Step #1	16:57:49	7500	18270	818.3	840.0
4	Step #2	16:59:01	6000	16000	787.1	810.0	5	Step #3	17:00:16	4500	11900	785.8	810.0
6	Step #4	17:02:27	3000	9250	787.8	810.0	7	Step #5	17:03:23	1500	6000	785.4	810.0
8	Step #6	17:03:59	100.0	2200	786.2	810.0	9	Copilot Disabled	17:05:14	100.0	2200	710.6	810.0

Figure 5—Control System Gripper Test

The same behavior was evidenced in the reel control system. By observing the decrease of the set point in reel hydraulic pressure and the corresponding applied pressure, while decreasing the weight and depth, as evidenced in events 4 and 5 of Fig. 6. We could also observe that throughout the test, hydraulic pressure control was stable after every system adjustment for both gripper and reel hydraulic control systems. With these satisfactory results, customer design of service and system test approvals, contingency plans, risk analysis and controls applied in place, the system was ready to be activated in operation.



Intersection		Time	CD	CPW	CRBT	CRPS
2	Copilot Enabled	16:57:09	7700	20000	1202	1200
3	Step #1	16:57:49	7500	18270	1201	1200
4	Step #2	16:59:01	6000	16000	1149	1150
5	Step #3	17:00:16	4500	11900	1049	1050
6	Step #4	17:02:27	3000	9250	1049	1050
7	Step #5	17:03:23	1500	6000	1050	1050
8	Step #6	17:03:55	100.0	2200	1051	1050
9	Copilot Disabled	17:05:14	100.0	2200	917.4	1050

Figure 6—Control System Reel Test

Operation Execution

As planned, in location the CT was rigged up with operation and safety controls in place and through well intervention job, the control system was activated during RIH for both gripper and reel circuits, and the system initialized with a traction (gripper) pressure of 800 psi. As the CT string advanced in depth, the control model continuously evaluated stress conditions according to sensor readings, CT string parameters, and adjusted the injector traction force accordingly.

From 150 ft to target depth (7700 ft), the system established a constant gripper pressure of 800 psi, even during CT weight checks performed in pull tests, which was the necessary force to hold the CT string safely and without overstressing during the complete well access, as illustrated in Fig. 7.

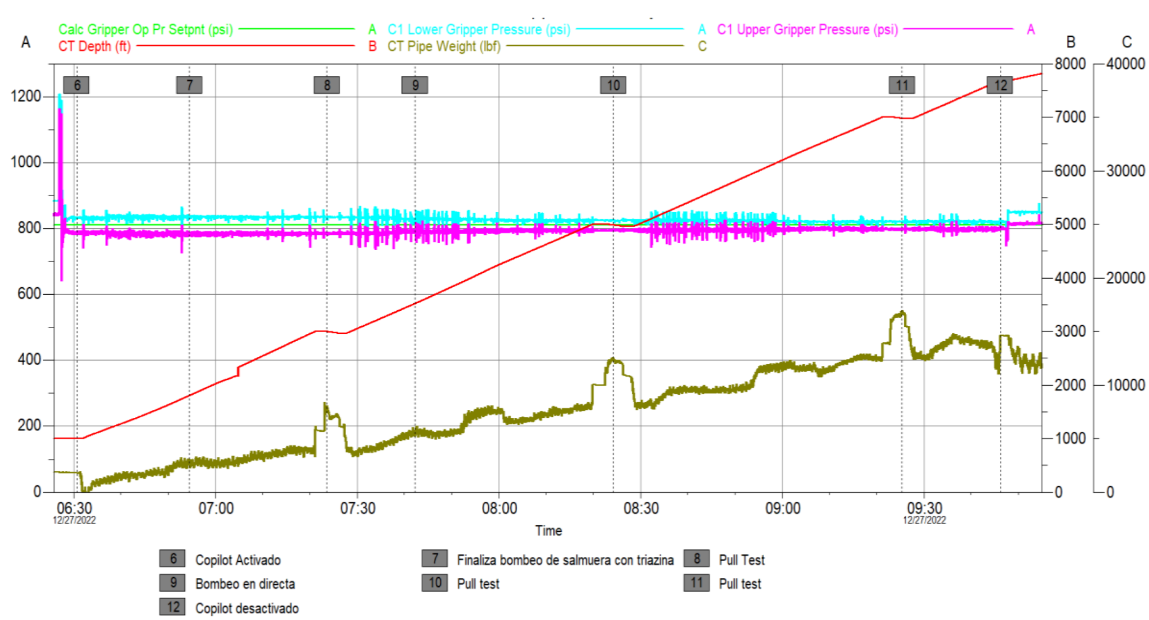


Figure 7—RIH Gripper Control System

At the same time, the reel control system initially controlled the hydraulic pressure to around 800 psi. However, it was observed during the pull test (events 8, 10, and 11) that the hydraulic pressure increased to a maximum of 1200 psi due to the additional tension force required to spool the CT string correctly. Following the pull test, went back to the last hydraulic pressure setpoint due to a change in CT direction. It was also evident that as the CT Wall Thickness and CT string weight increased, the control system should adjust the hydraulic reel pressure, as shown in Fig. 8.

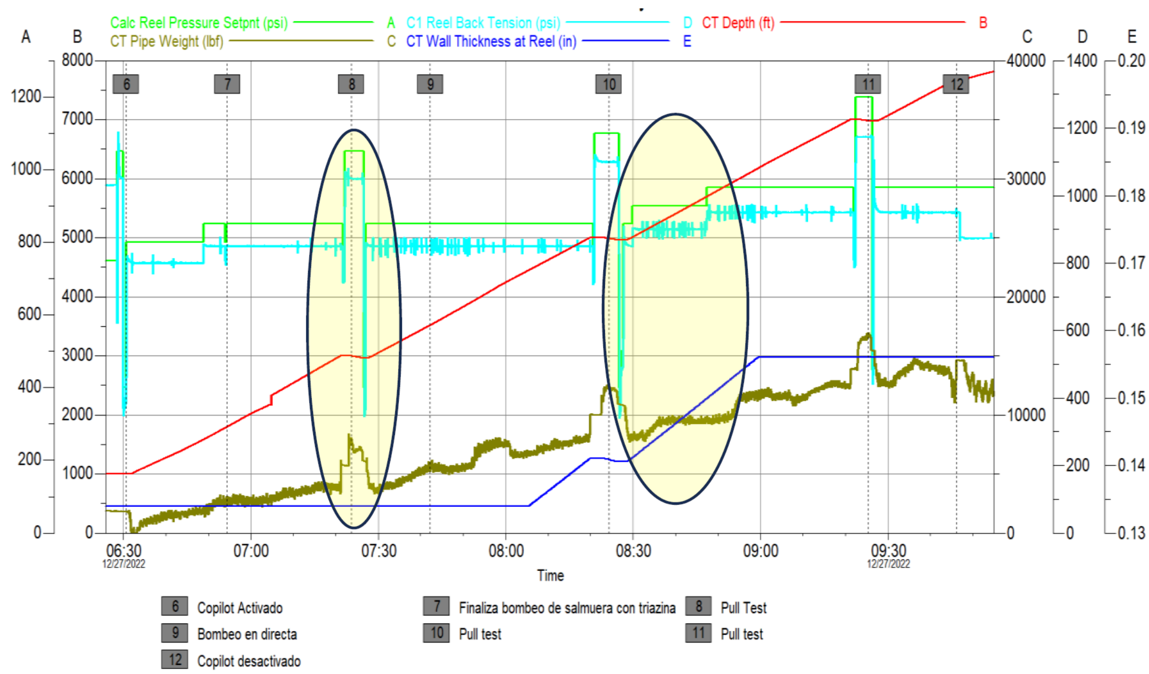


Figure 8—RIH Reel Control System

Once at the target depth, the control system was disabled, and the mechanical cleaning operation was performed normally in the interest intervals, according to job design with control system in manual mode. When it came time to pull the CT string back to the surface, after operation was completed, the control system was enabled again. Both transitions from automatic to manual and from manual to automatic were performed without compromising the job execution plan, however, in background, systems were always indicating the optimal operational values, so the operator always knew the appropriate values and had the possibility to apply them to hydraulic circuits.

During CT POOH, the gripper control system adjusted the corresponding hydraulic pressure according to CT string weight changes, ensuring that the appropriate force was applied at all times. It was observed that hydraulic gripper pressure tends to follow weight behavior indicating dynamic adjustment and stable pressure control capability, as shown in Fig. 9.

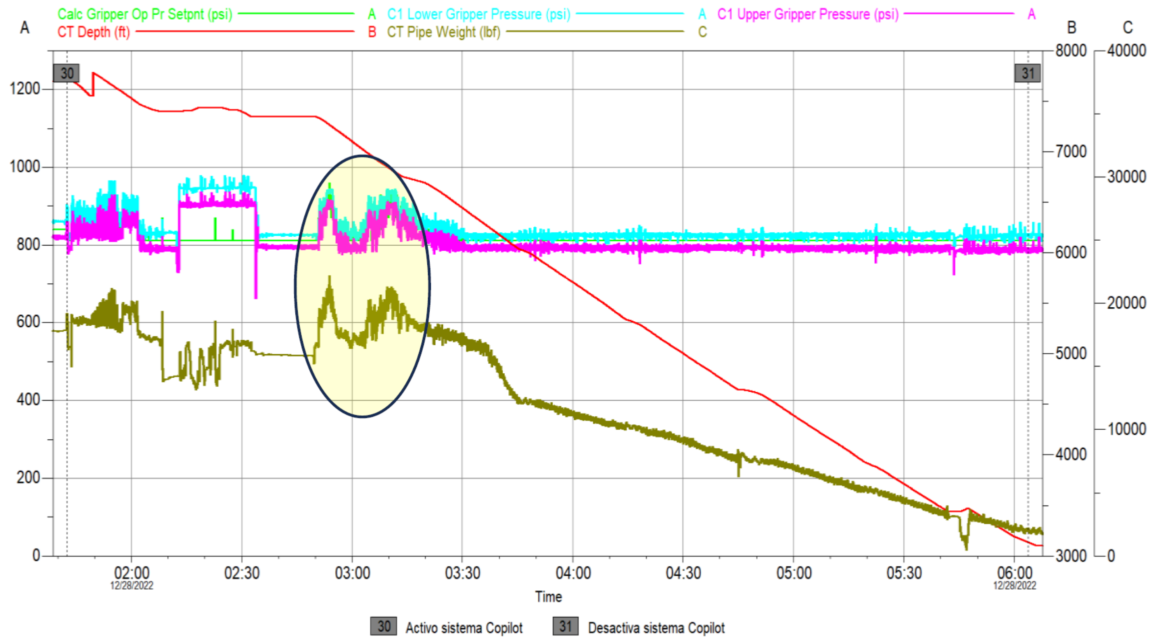


Figure 9—POOH Gripper Control System

In parallel, the reel tension control system adjusted and controlled the hydraulic pressure required to maintain a controlled back-tension, due to actual wall thickness and CT weight, to spool correctly the CT string, as observed in Fig. 10. For both gripper and reel control systems a stable pressure control was observed as set points were moving due to dynamic change in CT string parameters and reading sensors. 200 ft before CT arrived at surface, control system was finally disabled, and the operation was manually completed upon arrival CT string at surface. The job was successfully completed from an operational execution and control system performance standpoint, without any event that compromises security or service quality.

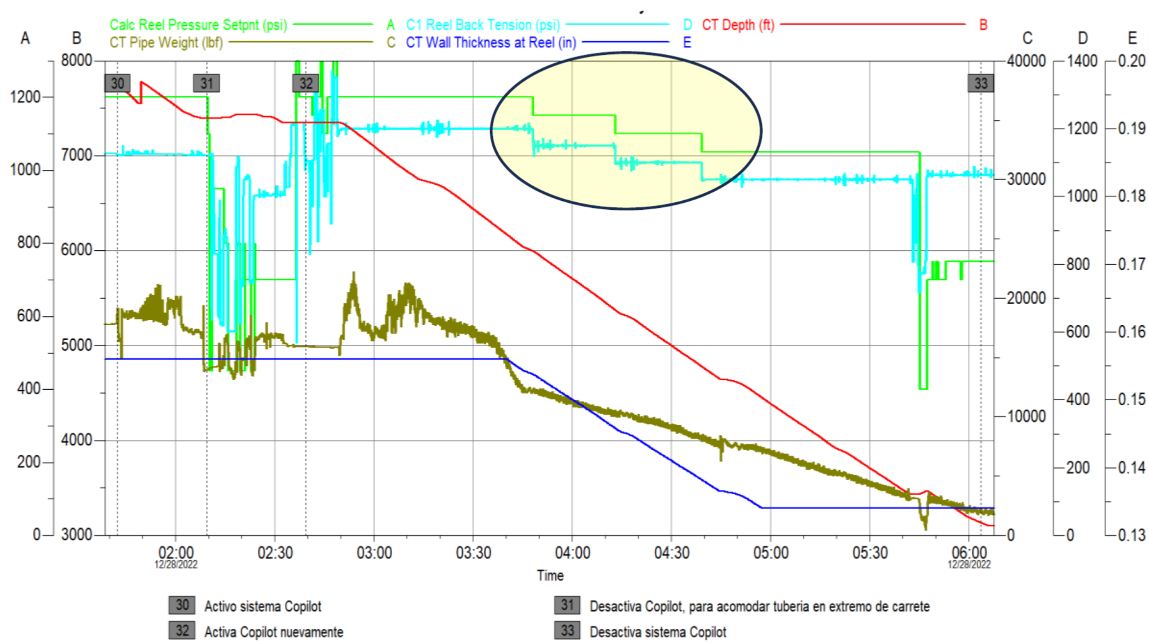


Figure 10—POOH Reel Control System

Operator Feedback and System Performance

Operators reported that the system provided valuable assistance without interfering with manual control and smooth operation of the control system. Key observations included:

- **Improved consistency** in force application across varying CT and well conditions updating control setpoints permanently.
- **Reduced operator workload**, assisting the CT operator by notifying of system adjustments in real time, without affecting the operator's autonomy to decide whether to operate in automatic or manual mode.
- **No operational delays or safety incidents** were attributed to the control system, transition between automatic and manual mode was executed as planned.

Discussion

The field implementation of the semi-autonomous control system demonstrated the practical viability of integrating real-time modeling and adaptive hydraulic control into CT operations. The results highlight several key insights regarding system performance, operator interaction, and the broader implications for CT automation.

System Responsiveness and Accuracy

The system effectively responded to dynamic changes in tubing conditions, such as variations in wall thickness, weight, internal and external pressure, and CT direction. By continuously solving the physics-based model and adjusting hydraulic pressures setpoints in real time, the system maintained not only automates force application but also enhances situational awareness for operators keeping stress levels within safe operational limits.

Operator Support and Autonomy

One of the most significant outcomes was the system's ability to assist operators without compromising their control. The hybrid control approach—combining automation with manual override—allowed operators to retain decision-making authority as well as make informed decisions without being overwhelmed by data, while benefiting from real-time alerts, visualizations, and automated adjustments. This balance is crucial in field environments where human expertise remains indispensable, especially during unexpected events or complex well conditions.

Operational Efficiency and Safety

The automation of injector traction and reel tension control contributed to improving the application of hydraulic gripping and tension forces, according to CT conditions. These improvements not only enhance operational efficiency but also reduce the risk of equipment damage and safety incidents. The system's ability to maintain consistent force application across varying depths and tubing conditions suggests potential for reducing CT fatigue accumulation and extending service life.

Lessons Learned

A significant portion of the effort in implementing the Gripper and Reel control system was dedicated to understanding and addressing the following key aspects:

- **Transition between manual and automatic modes** – The system allows operators to switch between manual and automatic control of the hydraulic circuits at any time. Ensuring seamless and safe transitions between these states require careful controller design and validation.

- **Risk Assessment** – A detailed Failure Modes and Effects Analysis (FMEA) was conducted to identify potential failure scenarios and corresponding mitigations. The analysis focused on defining safe fallback states for the system in cases such as sensor signal loss, unexpected actuator response times, and delays in settling to new setpoints.
- **Control System Response Tuning** – PID gains were adjusted to ensure that the control system met desired response time criteria while maintaining stability and precision.
- **Model Development** – Physics-based models were developed from first principles to calculate the required pressure setpoints for both Gripper and Reel operations. These models were validated using real job data to ensure accuracy and reliability."

Broader Implications
This work represents a step towards more intelligent and autonomous CT systems. By embedding real-time decision-making capabilities into the control loop, the system lays the groundwork for future developments in fully automated well intervention technologies. The approach also aligns with broader industry trends toward digitalization, predictive maintenance, and data-driven optimization in oilfield operations.

Conclusion

This paper presented implementation and field validation of a semi-autonomous control system for CT operations, aimed at extending tubing service life and enhancing operational safety. By integrating real-time physics-based modeling with electro-hydraulic control, the system dynamically adjusted injector traction and reel tension forces based on continuously updated surface sensor data and tubing mechanical properties. This approach represents an advancement in CT automation, delivering practical value through model-driven force application, thereby extending CT life.

The field trial conducted in the Llanos Orientales region of Colombia demonstrated the system's ability to maintain optimal force application during both CT RIH and POOH operations. The system successfully responded to variations in tubing geometry, internal pressure, external pressure, and weight, while providing real-time feedback and preserving operator autonomy to decide which system wants to activate in automatic and for how long.

Key outcomes include improved consistency in force control, reduced operator workload by notifying the appropriate hydraulic control setpoints in real time regardless control system was in manual or automatic mode, and enhanced protection of the CT string from mechanical overstress. The use of Von Mises equivalent stress as a control reference proved effective in maintaining structural integrity and preventing fatigue-related failures. The system's ability to maintain stress levels within safe limits contributed to preserving CT integrity throughout the operation.

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